

FARHAN KHAN

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SUMMARY

Final-year CS student with production-grade systems across full-stack, AI/ML, and real-time platforms. Built and shipped five end-to-end applications spanning WebSocket communication, on-chain voting, clinical AI, and industrial computer vision. Interned at Tata Steel and Tata Motors, building ERP-integrated AI systems and ensemble attrition models.

EDUCATION

Symbiosis Institute of Computer Studies & Research, Pune

Expected Jun 2027

BCA. (Hons) with Research | Symbiosis International University

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C++

Frontend: React, Next.js, Tailwind CSS, Vite, HTML/CSS

Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets

Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM

AI / ML: PyTorch, Scikit-learn, XGBoost, SHAP, YOLOv8, Pandas, OpenCV

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, Vercel

EXPERIENCE

Tata Steel | AI/ML Engineer Intern

Jun 2025 – Jun 2025 | Jamshedpur

- Built an Auto GRN Automation System integrating YOLOv8-based defect detection with SAP ERP workflows, eliminating manual goods receipt entry for steel coil inspection.
- Reduced manual QC intervention by automating the image-to-ERP validation pipeline; the system processed incoming material receipts end-to-end with AI-based anomaly checks.
- Designed REST API layer between Flask inference server and ERP integration module; manually annotated training dataset with bounding-box labels, achieving mAP@50 of 87.3%.

Hybionics | Machine Learning Intern

Mar 2026 – Apr 2026 | Remote

- Processed IMU sensor datasets (FallAIIID, SisFall) for fall/non-fall binary classification on a smart prosthetics platform; identified and resolved inverted label encoding mismatch between datasets that would have degraded model performance.
- Defined validation parameters, sanity checks, and labelling conventions for gait and fall event classification across multi-sensor modalities.

Tata Motors | Data Science & AI/ML Intern

Apr 2026 – Jun 2026 | Jamshedpur

- Built a stacked ensemble model (XGBoost + LightGBM + CatBoost) for employee attrition prediction with ~89% accuracy and advanced feature engineering on HR demographic data.
- Delivered FastAPI-served inference endpoint and React dashboard for HR analytics, enabling real-time attrition risk scoring per employee.

PROJECTS

ConnectX – Real-Time Communication Platform | Node.js, React, MongoDB, Socket.IO, JWT

- Built a full-stack messaging platform with WebSocket-based real-time communication, JWT auth, voice messaging, file sharing, and scheduling — shipped as a single-deploy app.
- Integrated AI-powered link previews and external API hooks; designed a 22-feature expansion roadmap with complete test coverage.

BlockVote – Decentralised Voting System | Solidity, Ethereum, React, Node.js, PostgreSQL, Hardhat

- Deployed Ethereum smart contracts for on-chain election management with wallet-based authentication; backend indexing service kept PostgreSQL read-replica in sync with chain events.
- Implemented tamper-proof vote verification with full audit trail; contract suite tested on Hardhat local network before Goerli testnet deploy.

NephroSense – CKD Risk Assessment Platform | React, Flask, XGBoost, SHAP

- Built a clinical decision support platform achieving good accuracy and recall on the UCI CKD dataset using XGBoost at a calibrated threshold of 0.213.
- Integrated SHAP explainability layer for per-patient feature attribution, PDF report generation, and bulk patient triage dashboard.

Sheet Metal Defect Detector | YOLOv8, Flask, React, REST APIs

- Trained YOLOv8 on a manually annotated steel surface dataset; deployed real-time inference via Flask REST API with React frontend.
- Achieved mAP@50 of 87.3% on held-out test set.

CERTIFICATIONS

AI for Business Professionals (HP) • Machine Learning with Python (Cognitive Class) • Data Analysis with Python (Cognitive Class) • AWS Cloud Essentials (AWS)